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Summer Internship Report

Home Automation Using ESP32

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Maven Silicon

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ABSTRACT

The Home Automation System using ESP32 aims to provide an efficient, low-cost, and user-friendly solution for remotely controlling and monitoring home appliances. The ESP32 microcontroller, equipped with built-in Wi-Fi and Bluetooth capabilities, acts as the central processing unit of the system. Various sensors such as the DHT sensor, LDR, and ultrasonic sensor are integrated to monitor environmental parameters like temperature, light intensity, and object proximity. Based on these sensor readings, the system can automatically control devices such as fans and lights in automatic mode, while also allowing manual control through a web interface or mobile device in manual mode. The project demonstrates how IoT technology can enhance comfort, improve energy efficiency, and reduce human effort by automating daily household operations. This system can be easily extended to support additional smart devices, making it a scalable and practical solution for modern smart homes.

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1 INTRODUCTION

1.1 Introduction

In today's fast-paced world, technology is advancing rapidly, transforming the way people live and interact with their surroundings. One such transformation is the concept of **home automation**, which involves the control and automation of lighting, appliances, heating, security systems, and more through electronic means.

The integration of the **Internet of Things (IoT)** has further enhanced home automation by enabling users to remotely control devices via smartphones or computers over the internet. This project, **"Home Automation using NodeMCU"**, explores the design and implementation of a low-cost, Wi-Fi-based smart home system using the NodeMCU (ESP32) microcontroller. The system allows users to remotely control electrical appliances like lights and fans through a mobile application such as Blynk or a custom interface.

The NodeMCU, with its built-in Wi-Fi capability and compatibility with cloud platforms, makes it an ideal choice for such IoT-based applications.

The importance of home automation is evident in various application scenarios:

- **Energy Efficiency:** Users can turn off unused appliances remotely, reducing energy consumption and electricity bills.
- **Elderly and Disabled Assistance:** Automation provides greater accessibility for individuals with mobility challenges, enabling them to control appliances without moving.
- **Security:** Smart home systems can be extended with sensors and surveillance modules to enhance home security through remote monitoring and alert systems.
- **Convenience:** Everyday tasks such as turning on the water heater or lighting can be scheduled or triggered automatically, improving comfort and lifestyle.
- **Scalability:** Systems like this are modular and can be easily extended to control multiple devices, rooms, or even entire buildings.

1.2 Problem Statement

Despite technological advancements, many households still rely on manual switching of electrical appliances, which leads to energy wastage and inconvenience. Often, users forget to turn off lights or fans, resulting in unnecessary power consumption.

Additionally, individuals with physical limitations face difficulties in accessing switches, while remote control of appliances remains inaccessible in traditional systems. These challenges highlight the need for a smart, connected, and energy-efficient solution that enables automation and remote access.

The problem addressed by this project is to design and implement an IoT-based home automation system that:

- Provides both **manual and automatic control** of household appliances.
- Operates **remotely via Wi-Fi** using a smartphone interface.
- Offers a **low-cost, scalable, and user-friendly** solution.

1.3 Objectives

The main objectives of the **Home Automation using NodeMCU** project are:

1. **To design a cost-effective IoT-based home automation system** using the NodeMCU (ESP8266) microcontroller for controlling home appliances via Wi-Fi.
2. **To enable remote access and control** of electrical devices such as lights and fans using a smartphone app (e.g., Blynk).
3. **To improve energy efficiency** by allowing users to monitor and switch off appliances when not in use.
4. **To enhance safety and comfort** through the potential integration of sensors for fire, gas leakage, or intrusion detection.
5. **To provide flexibility and scalability** for future expansion to multiple rooms or devices.
6. **To give hands-on experience** in embedded systems, IoT communication, and cloud interfacing for students and researchers.

In summary, this project not only demonstrates the potential of IoT-based smart home systems but also contributes to energy conservation, accessibility, and convenience in modern living environments.

2 PROJECT SPECIFICATIONS

2.1 Functional Requirements

- This project controls two home appliances, a light bulb and a fan connected to a 230v supply through relays.
- The controller board is connected to the home WiFi network.
- The fan and bulb are controlled by a mobile connected to home WiFi network.
- Auto or Manula Control mode can be selected on the mobile phone.

Mode Of Operation

- Auto Mode
  - When a person enters the room and if no day light in the room then the bulb will be switch on.
  - When the person moves out,the bulb will switch off.
  - When the temperature in the room goes beyond a threshold the fan will start and temperature goes below the threshold the fan will stop.
- Manual Mode
  - The fan and bulb can be switched on and off through the mobile phone connected to the same home network by browsing to the appropriate IP address.
  - The bulb and fan cannot be controlled manually if it is in auto mode.

2.2 Non-Functional Requirements

Performance Requirements:
Automatic mode:

- The bulb should switch on after the entry of uman in the room within 1 minute.

- Fan should switch on as soon as the threshold is detected within 1 minute.

Manual Control mode:

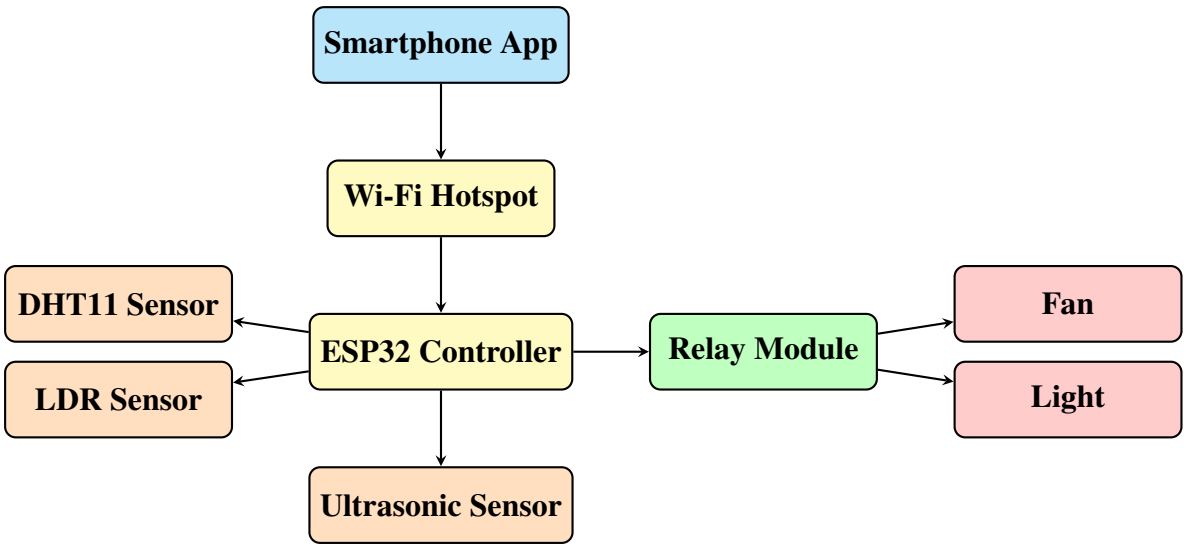
- The bulb should switch on within 1 minute of pressing the button on the mobile phone.
- Fan should switch on within 1 minute of pressing the button on the mobile phone.

Environment Requirements

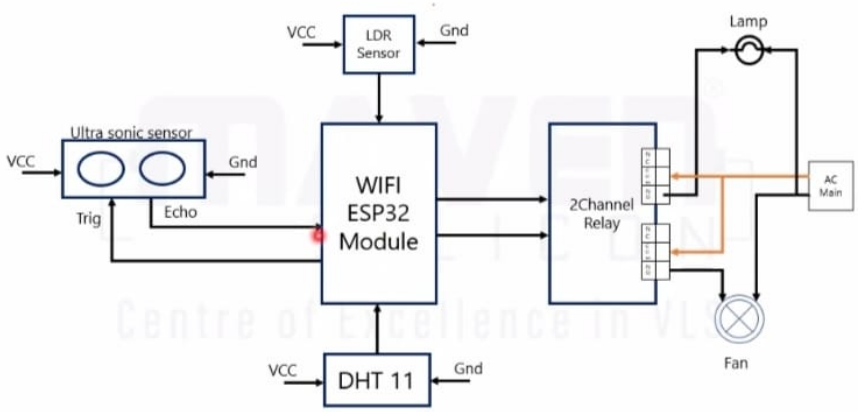
- The system should work in the temperature range 0 degree centigrade to 60 degree centigrade.

3 HARDWARE ARCHITECTURE

3.1 Block Diagram



3.2 Schematic Diagram



Selection of Microcontroller

Microcontroller Board Selection: ESP32 (NodeMCU) The ESP32-based NodeMCU development board was chosen due to its balance of performance, cost, and built-in Wi-Fi capability. The key factors that led to this decision are:

- **Built-in Wi-Fi:** Ideal for IoT and home automation applications where remote control or monitoring is required.
- **GPIO Availability:** Sufficient number of GPIO pins to interface with multiple sensors and actuators.
- **Arduino Compatibility:** Easily programmable via the Arduino IDE with a vast number of community-supported libraries.
- **Cost-Effective:** Affordable and widely available in the market, making it ideal for prototyping and educational purposes.
- **Compact Size:** Fits well in small enclosures and embedded setups.

Justification for Selected Peripheral Devices

The selection of peripheral devices was based on their relevance, cost-efficiency, and ease of integration with the ESP32 microcontroller.

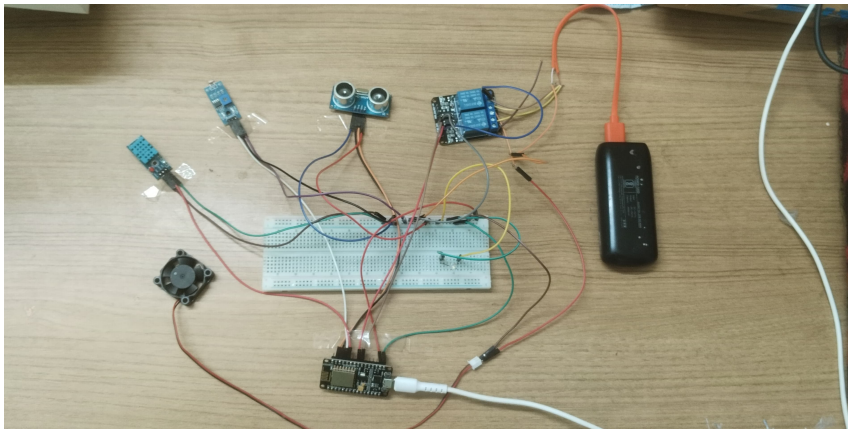
- **DHT11 Sensor:** Chosen for its ability to provide both temperature and humidity readings with sufficient accuracy for home automation applications. It is widely supported and easy to interface using the DHT.h library.
- **LDR Sensor:** Selected to detect ambient light levels, enabling automatic switching of lights during low-light conditions. It is inexpensive and works well as a digital sensor with basic thresholding logic.
- **Ultrasonic Sensor (HC-SR04):**Utilized to detect human presence by measuring distance. It provides non-contact and accurate measurement, which helps in determining occupancy of the room to control fan/light.
- **Relay Module (2-Channel):** Used to interface AC devices (fan and light) with the microcontroller. The relay acts as an electrically operated switch to safely control high-voltage appliances using low-voltage GPIO pins.
- **Fan and Light (AC Loads):** Represent real-world applications of automation. These appliances are commonly found in homes and are ideal candidates for demonstrating automatic and manual control features.

- **ESP8266 (NodeMCU):** Selected for its built-in Wi-Fi capabilities, affordability, and availability of GPIO pins to connect multiple sensors and relays. It also supports web server functionality for manual control.

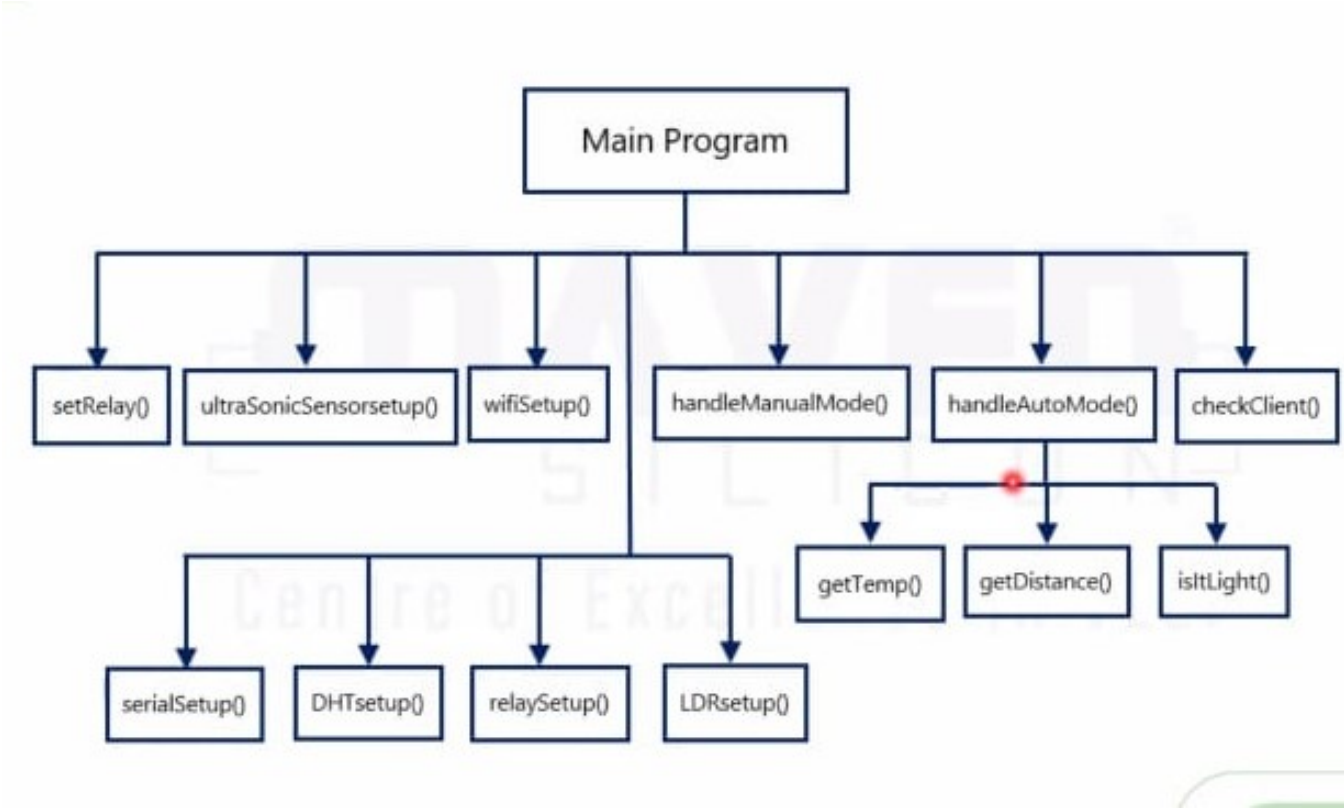
3.3 List of hardware components

Table 3.1: List of Materials

Name	Description	Quantity
ESP32 Microcontroller	Main control unit with built-in Wi-Fi and GPIO for sensor and relay interfacing.	1
Relay Module (2-Channel)	Used to control AC devices like fan and light from the microcontroller.	1
DHT11 Sensor	Temperature and humidity sensor with digital output.	1
LDR Sensor	Used to detect ambient light levels. It allows automatic control of lighting based on environmental brightness.	1
HC-SR04 Ultrasonic Sensor	Sensor for proximity detection. It helps in triggering automation only when a user is nearby, conserving energy.	1
DC Fan	Output device controlled by the relay, turns ON/OFF based on temperature or user input.	1
Light (Bulb or LED Lamp)	Output device controlled by the relay, turns ON/OFF based on light level or user input.	1
Wires, Breadboard, Power Supply	For circuit prototyping, power distribution, and sensor connections.	As required



4 SOFTWARE ARCHITECTURE



Flowchart of the Home Automation Program

5 PSEUDOCODE

```
1
2 BEGIN
3
4   // --- Initialization Phase ---
5   Initialize Serial Communication at 115200 baud rate
6   Connect to Wi-Fi using SSID and Password
7   Wait until Wi-Fi connection is successful
8   Start the web server
9
10  Initialize DHT sensor for temperature reading
11  Initialize LDR sensor for light detection
12  Initialize Ultrasonic sensor for distance measurement
13  Initialize relay pins for Light and Fan control
14
15  // --- Main Execution Loop ---
16  WHILE (true) DO
17       Check if any client request is available
18
19      IF (mode == AUTO) THEN
20          // --- Automatic Mode ---
21          temperature     Read temperature from DHT sensor
22          distance     Measure distance using Ultrasonic
               sensor lightStatus     Read LDR sensor
23
24          IF (distance < thresholdDistance) THEN
25              IF (temperature > temperatureLimit) THEN
26                  Turn ON Fan
27              ELSE
28                  Turn OFF Fan
29              ENDIF
```

```
30
31              IF (lightStatus indicates DARK) THEN
32                  Turn ON Light
33              ELSE
34                  Turn OFF Light
35              ENDIF
36
37          ELSE
38              Turn OFF Fan
39              Turn OFF Light
40          ENDIF
41
42      ELSE
43          // --- Manual Mode ---
44          IF (client is connected) THEN
45              IF (client sends data) THEN
46                  command     Read client input
47
48                  IF (command == "fanon") THEN
49                      Turn ON Fan
50                  ELSE IF (command == "fanoff") THEN
51                      Turn OFF Fan
52                  ELSE IF (command == "bulbon") THEN
53                      Turn ON Light
54                  ELSE IF (command == "bulboff") THEN
55                      Turn OFF Light
56                  ENDIF
57              ENDIF
58          ENDIF
59      ENDIF
60  ENDWHILE
61  END
```

6 SOURCE CODE

ESP32 Home Automation Code

```
#include <ESP8266WiFi.h>
#include "DHT.h"

#define DHTPIN D4 // GPIO2
#define DHTTYPE DHT11
#define LDR_PIN D3 // GPIO0
#define TRIG_PIN D1 // GPIO5
#define ECHO_PIN D2 // GPIO4
#define RELAY_LIGHT D5 // GPIO14
#define RELAY_FAN D6 // GPIO12

DHT dht(DHTPIN, DHTTYPE);
WiFiServer server(80);
bool isManualMode = false;
uint8_t previousRelayLight = HIGH;
uint8_t previousRelayFan = HIGH;
const char* ssid = "tejaswi";
const char* password = "12345678";

void serialSetup() {
  Serial.begin(115200);
  delay(10);
  Serial.println();
}
void DHTsetup() {
  dht.begin();
}
void LDRsetup() {
  pinMode(LDR_PIN, INPUT);
}
void ultrasonicSetup() {
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
```

```
}
void relaySetup() {
  pinMode(RELAY_LIGHT, OUTPUT);
  pinMode(RELAY_FAN, OUTPUT);
  digitalWrite(RELAY_LIGHT, HIGH);
  digitalWrite(RELAY_FAN, HIGH);
}
void wifiSetup() {
  Serial.print("Connecting to ");
  Serial.println(ssid);
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nWiFi connected");
  Serial.print("IP address: ");
  Serial.println(WiFi.localIP());
  server.begin();
}
float getTemperature() {
  delay(1000);
  float t = dht.readTemperature();
  Serial.print("Temp: ");
  Serial.println(t);
  return t;
}
bool isItDark() {
  int state = digitalRead(LDR_PIN);
  Serial.println(state == HIGH ? "It's dark" : "It's light");
  return state == HIGH;
}
float getDistance() {
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
  long duration = pulseIn(ECHO_PIN, HIGH);
  float distance = duration * 0.034 / 2;
  Serial.print("Distance: ");
  Serial.println(distance);
  return distance;
}
void setRelay(uint8_t pin, uint8_t value) {
```

```
  value = !value;
  if (pin == RELAY_LIGHT && previousRelayLight != value) {
    digitalWrite(RELAY_LIGHT, value);
    previousRelayLight = value;
    Serial.println(value == LOW ? "Light ON" : "Light OFF");
  }
  if (pin == RELAY_FAN && previousRelayFan != value) {
    digitalWrite(RELAY_FAN, value);
    previousRelayFan = value;
    Serial.println(value == LOW ? "Fan ON" : "Fan OFF");
  }
}
void handleAutoMode() {
  float distance = getDistance();
  float temp = getTemperature();
  bool dark = isItDark();

  if (distance <= 10) {
    if (temp >= 20) setRelay(RELAY_FAN, LOW);
    else setRelay(RELAY_FAN, HIGH);

    if (dark) setRelay(RELAY_LIGHT, LOW);
  } else {
    setRelay(RELAY_LIGHT, HIGH);
    setRelay(RELAY_FAN, HIGH);
  }
}
void handleClient(WiFiClient client) {
  String request = "";
  while (client.connected()) {
    if (client.available()) {
      char c = client.read();
      request += c;
      if (c == '\n') {
        if (request.indexOf("GET /?lightOn") >= 0) setRelay(RELAY_LIGHT,
    LOW);
        if (request.indexOf("GET /?lightOff") >= 0) setRelay(RELAY_LIGHT,
    HIGH);
        if (request.indexOf("GET /?fanOn") >= 0) setRelay(RELAY_FAN, LOW);
        if (request.indexOf("GET /?fanOff") >= 0) setRelay(RELAY_FAN, HIGH)
    ;
        if (request.indexOf("GET /?manualOn") >= 0) isManualMode = true;
        if (request.indexOf("GET /?manualOff") >= 0) isManualMode = false;

        client.println("HTTP/1.1 200 OK");
```

```
121         client.println("Content-Type: text/html");
122         client.println();
123         client.println("<html><head><title>ESP8266 Control</title></head><
    body>");
124         client.println("<h1>ESP8266 Home Automation</h1>");
125         client.println("<p>Light Control: <a href='/?lightOn'><button>ON</
    button></a> <a href='/?lightOff'><button>OFF</button></a></p>");
126         client.println("<p>Fan Control: <a href='/?fanOn'><button>ON</
    button></a> <a href='/?fanOff'><button>OFF</button></a></p>");
127         client.println("<p>Manual Control: <a href='/?manualOn'><button>ON
    </button></a> <a href='/?manualOff'><button>OFF</button></a></p>");
128         client.println("</body></html>");
129         break;
130       }
131     }
132   }
133   delay(1);
134   client.stop();
135 }
136 void setup() {
137   serialSetup();
138   DHTsetup();
139   LDRsetup();
140   ultrasonicSetup();
141   relaySetup();
142   wifiSetup();
143 }
144 void loop() {
145   WiFiClient client = server.available();
146   if (client) handleClient(client);
147   if (!isManualMode) {
148     handleAutoMode();
149   }
150   delay(2000);
151 }
```

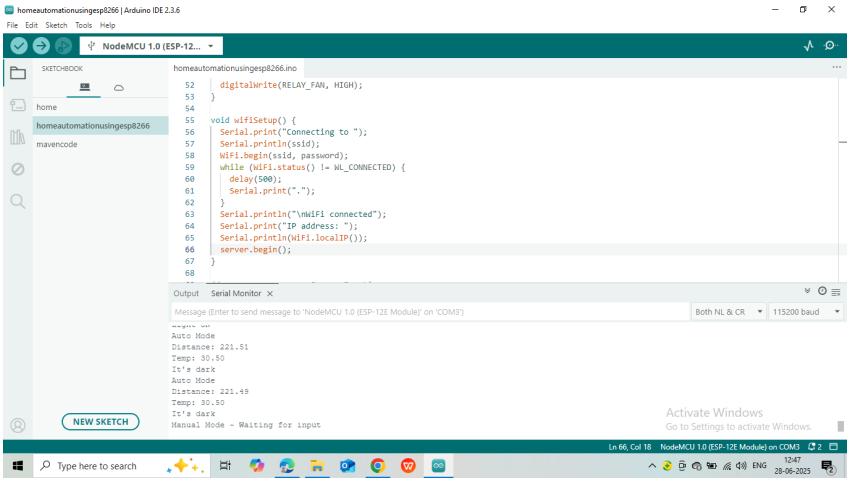
Listing 6.1: ESP8266 Home Automation Full Code

7 TESTING AND RESULTS

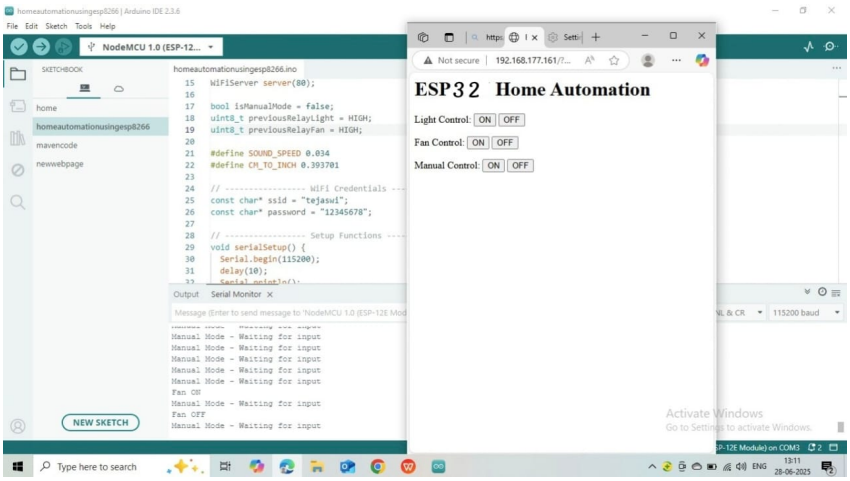
Table 7.1: Test Cases for ESP32 Home Automation System

S.No	Input	Expected Output	Mode	Result
1	Valid WiFi SSID and password	ESP connects and prints IP address	Auto	Pass
2	Temperature from DHT11 sensor	Displays temperature in serial monitor	Auto	Pass
3	Light condition detected by LDR	Detects whether it is light or dark	Auto	Pass
4	Person placed 10 cm from ultrasonic sensor	Returns distance around 10 cm	Auto	Pass
5	Temp ¿ 28°C, Person present	Fan turns ON automatically	Auto	Pass
6	Dark room, Person present	Light turns ON automatically	Auto	Pass
7	User clicks “Light ON” button on web page	Light turns ON	Manual	Pass
8	User clicks “Fan OFF” button on web page	Fan turns OFF	Manual	Pass
9	No person detected, light ON	Light turns OFF automatically	Auto	Pass
10	Client sends malformed HTTP request	No response or ignored	Manual	Pass

7.1 RESULTS



Automode



Manual mode: Buttons are provided to manually switch ON/OFF the light and fan. The user can toggle between Auto and Manual mode through the same interface.



WebInterface:A minimal HTML page is served by the ESP32 containing control buttons.

8 FUTURE SCOPE

The Home Automation System developed using the ESP832 microcontroller offers a strong foundation for smart living. However, there is vast potential for further enhancement and scalability. Some of the key areas of future scope are:

- **Voice Assistant Integration:** Integration with popular voice assistants like Amazon Alexa or Google Assistant to control devices through voice commands.
- **Mobile App Development:** Developing a dedicated mobile application (Android/iOS) for more user-friendly and intuitive control over appliances, replacing the need for web-based interfaces.
- **Cloud-Based Data Logging:** Storing temperature, humidity, and motion data in the cloud for historical analysis and predictive automation.
- **Machine Learning Integration:** Incorporating AI and ML algorithms to predict user behavior and automate device operation based on past usage patterns.
- **Energy Monitoring and Optimization:** Adding energy sensors to monitor power consumption of connected devices and recommend ways to reduce electricity usage.
- **Security Enhancements:** Implementing user authentication and encrypted communication for enhanced security, especially in IoT-based systems.
- **Multi-Device Synchronization:** Extending the system to manage and synchronize multiple rooms or zones, enabling full-house automation.
- **Sensor Expansion:** Integrating additional sensors like gas leak detectors, fire alarms, and smart meters to widen the scope of automation and safety

9 REFERENCES

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